Prayush B Menon

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OBJECTIVE

Embedded Systems & Firmware Engineer with an ECE background and hands-on expertise in bare-metal C development on PIC microcontrollers. Proficient in serial communication protocols (CAN, UART, SPI, I2C), Linux systems programming (POSIX, IPC, signals), memory-efficient data structures, and GDB debugging. Experienced in low-level firmware architecture, hardware-software integration, and software test automation.

TECHNICAL SKILLS

Languages: C (C99/C11), Embedded C, C++, Java, Bash/Shell Scripting
Microcontrollers & Protocols: PIC18F4580, PIC16F877A, CAN, UART, I2C, SPI, ADC, EEPROM, RTC
Systems & Concepts: POSIX System Calls, IPC (Pipes, Signals), Multiprocessing, Memory Management, Bit Manipulation, FSM
Developer Tools & OS: Linux (Fedora/Ubuntu), GDB, Make, CMake, Git, Neovim, Vim, Postman, Burp Suite
Testing & Infrastructure: Selenium WebDriver, CI/CD Performance Monitoring (Lighthouse), Test Automation, JIRA

PROJECTS

Minishell (Unix Command Shell) GitHub

C, Linux Systems Programming, POSIX APIs, IPC, Signals, Process Management

- Built a custom POSIX Unix command shell in C supporting command execution, dynamic prompt customization, and multi-stage pipelining (N-pipes) via `pipe()` and `dup2()` file descriptor redirection.
- Engineered asynchronous signal handlers (`SIGINT`, `SIGTSTP`, `SIGCHLD`) and a linked-list process job control subsystem to manage suspended and background jobs with `waitpid()`.

Car Black Box (Event Data Recorder) GitHub

Embedded C, PIC16F877A, I2C, UART, ADC, EEPROM (AT24C04), RTC (DS1307), CLCD

- Engineered bare-metal firmware on a PIC16F877A microcontroller to log vehicle speed, gear state, and DS1307 RTC timestamps into external AT24C04 EEPROM via bit-level I2C protocol routines.
- Implemented a non-volatile circular buffer for event data retention across power resets, along with bare-metal drivers for ADC speed potentiometer sampling, matrix keypad debouncing, CLCD UI, and UART PC telemetry export.

Car Dashboard (Automotive CAN Simulator) GitHub

Embedded C, PIC18F4580, CAN Protocol, CLCD, Embedded Systems

- Configured PIC18F4580 registers and interrupt service routines to transmit real-time vehicle telemetry frames (speed, RPM, gear status) across an automotive CAN bus.
- Engineered state-driven display logic on a receiving CLCD ECU to filter, unpack, and render incoming CAN network frames in real time.

Lexical Analyzer for C GitHub

C, Lexical Analysis, Tokenization, Binary Search, File I/O

- Developed a stream-based lexical tokenizer in C to extract and classify tokens (keywords, identifiers, literals, operators) using $O(\log N)$ binary search lookup tables over sorted symbol arrays.
- Implemented character stream parsing state logic to strip preprocessor directives and comments while enforcing syntax validation, including bracket balance tracking (`()`, `{}`, `[]`) and literal verification.

Arbitrary Precision Calculator (APC) GitHub

C, Data Structures, Algorithms, Dynamic Memory Allocation

- Engineered an arbitrary-precision arithmetic engine in C using doubly linked lists to perform mathematical operations on integers of arbitrary size, overcoming standard 64-bit integer limits.
- Implemented base-10 addition, subtraction, multiplication, and division algorithms with carry/borrow management, sign handling, and zero-memory-leak dynamic allocation routines.

Image Steganography (LSB) GitHub

C, Bit Manipulation, Binary File I/O, Image Processing

- Implemented a binary file processing utility in C to embed and decode encrypted secret messages within 24-bit BMP images using Least Significant Bit (LSB) substitution.
- Utilized bitwise masking and shifting across RGB pixel arrays to embed payload data streams while preserving the 54-byte BMP header structure and image fidelity.

MP3 Tag Reader & Editor GitHub

C, Binary File Parsing, ID3v2 Tags, File I/O

- Developed a command-line binary parser in C to read, extract, and modify ID3v2.3 metadata tags directly within raw MP3 audio streams.
- Parsed variable-length frame headers, managed byte offsets, and handled big-endian conversions to update metadata in-place without corrupting the underlying audio bitstream.

Source2HTML (C Source to HTML Converter) GitHub

C, Finite State Machine (FSM), Lexical Parsing, File I/O, HTML/CSS

- Engineered an event-driven C syntax highlighter using a multi-state Finite State Machine (FSM) to tokenize source code into typed lexical events (keywords, directives, literals, comments).
- Implemented an HTML generator featuring CSS token class mapping, HTML entity escaping (`<`, `>`, `&`), and CLI line-numbering flags.

EXPERIENCE

Senior Quality Engineer I July 2024 – January 2026

Fresh Prints Remote

- Built an automated frontend performance testing infrastructure with Lighthouse, reducing manual testing time by 80% through bi-weekly automated performance monitoring across environments.
- Led backend data migration validation for 7,000+ design records from Shopify to Directus CMS, developing custom validation scripts that ensured 100% data integrity and schema consistency.
- Identified critical P1 security vulnerabilities using Burp Suite to detect unauthorized access vectors, collaborating directly with development teams for immediate patch verification.

QA Automation Engineer June 2023 – July 2024

Cognizant Technology Solutions Bangalore, India

- Developed and maintained test automation frameworks using Java and Selenium WebDriver, standardizing regression testing across core application modules.
- Triaged and tracked software defects across agile sprint cycles, performing root-cause analysis to verify fixes and ensure release reliability.
- Reviewed technical requirements and system specifications to author automated test plans, expanding test coverage across end-to-end user workflows.
- Collaborated with engineering teams to optimize test execution pipelines, migrating manual validation suites into continuous automated test runs.

QA Engineer June 2022 – June 2023

Cognizant Technology Solutions Bangalore, India

- Executed functional and regression test suites across multi-tier applications, isolating defect root causes and reducing turnaround time for issue resolution.
- Performed REST API validation and endpoint testing using Postman, verifying payload schemas, HTTP status codes, and backend data flows.
- Authored and maintained comprehensive test suites covering product functional specifications, mitigating critical defect escapes into production releases.
- Scripted test automation routines to replace repetitive manual verification, accelerating regression testing cycles while maintaining consistent test repeatability.

EDUCATION & TRAINING

Emertxe Information Technologies Bangalore, India

Advanced Embedded Systems Training (PIC18F4580, C/C++, Protocols, Linux) 2025 – 2026

Amrita School of Engineering Kollam, Kerala

B.Tech in Electronics and Communications Engineering (CGPA: 8.1/10) June 2018 – August 2022